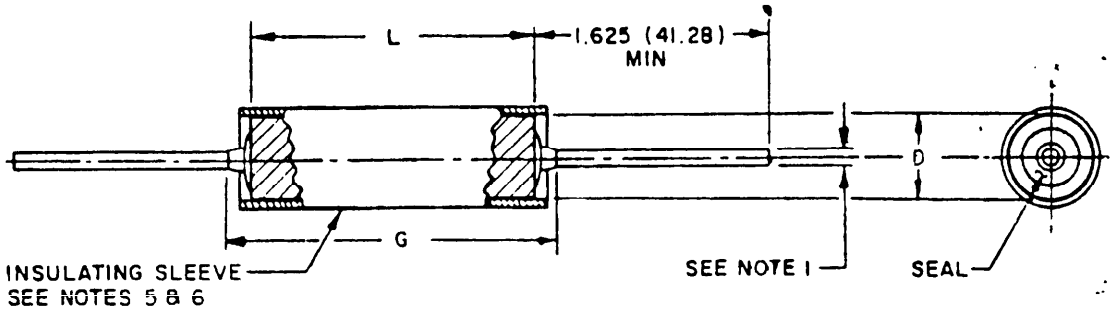
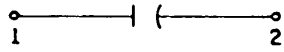
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, METALLIZED PLASTIC FILM DIELECTRIC, DIRECT AND ALTERNATING CURRENT, (HERMETICALLY SEALED IN METAL CASES), ESTABLISHED RELIABILITY, STYLES CHR01A, CHR01B, CHR01C, CHR01D, CHR01E, CHR01F, CHR01G, CHR01H, CHR01J, CHR01K, CHR01L, CHR01M, CHR01N, CHR01P, AND CHR01R (INSULATED)

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-39022.

INACTIVE FOR NEW DESIGN AFTER 31 OCTOBER 1973.
FOR NEW DESIGN, USE MIL-C-83421.





CIRCUIT DIAGRAM 1

NOTES:
1. Number 22 AWG wire for case diameters of .312 (7.92 mm) and less. Number 20 AWG wire for case diameters of .400 (10.16 mm) and .500 (12.70 mm). Number 18 AWG wire for case diameters of .562 (14.27 mm) and over.
2. See table I for additional dimensions.
3. Dimensions are in inches.
4. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
5. Insulating sleeve shall extend beyond the capacitor body but shall not exceed .031 (.79 mm) inch on either end. Insulating sleeve thickness shall not exceed .005 (.13 mm) inch.
6. Plastic insulating sleeve shall be transparent; marking shall be applied to the capacitor case.
7. Metric equivalents are in parentheses.

REQUIREMENTS:
Design and construction -
Dimensions and configuration: See figure and table I.
© Case material: Nonmagnetic. Seal headers may be magnetic.
© Sleeves: Heat shrink only.
Leads:
.500 inch and less case diameters - In accordance with MIL-C-39022, or type N2 of MIL-STD-1276.
.562 inch and over case diameters - Type C of MIL-STD-1276.
Capacitance (Cap.) value: See table I.
Cap. tolerance: See table I.
Rated voltage: DC and AC (see table I).
Operating temperature range: -55° to +125° C.
Failure rate level: L (5.0%), M (1.0%), P (0.1%), R (0.01%), and S (0.001%) (established at dc rated voltage at +100° C).

© denotes changes.

FSC 5910

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:15Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical characteristics, dimensions, and dash numbers.

Capacitance value (nom) (in μF)	Dimensions 1/ (in inches with mm in parentheses) L ±.030 (.76)	D +.020 (.51) -.000	G, max	Dash number 2/ Capacitance tolerance value (in %) ±0.25	±0.5	±1.0	±2.0	±5.0	±10.0	AC ratings (for sinusoidal operation from -55° to +100°C) 3/ 0 to .4 KHz Volts	Current (in A)	At 4 KHz Volts	Current (in A)	At 40 KHz Volts	Current (in A)
STYLES CHR01A, CHR01B, AND CHR01C - 30 VOLTS (DC RATING)															
0.001	0.500 (12.70)	0.170 (4.32)	0.700 (17.78)	-001-	-002-	-003-	-004-	-005-	-006-	22.0	.001	22.0	.001	22.0	.006
0.0012				-007-	-008-	-009-	-010-	-011-	-012-						.007
0.0015				-013-	-014-	-015-	-016-	-017-	-018-						.008
0.0018				-019-	-020-	-021-	-022-	-023-	-024-						.010
0.002				-025-	-026-	-027-	-028-	-029-	-030-						.011
0.0022				-031-	-032-	-033-	-034-	-035-	-036-						.012
0.0027				-037-	-038-	-039-	-040-	-041-	-042-				.002		.015
0.0033				-043-	-044-	-045-	-046-	-047-	-048-				.002		.020
0.0047				-049-	-050-	-051-	-052-	-053-	-054-				.003		.026
0.005				-055-	-056-	-057-	-058-	-059-	-060-				.003		.028
0.0068				-061-	-062-	-063-	-064-	-065-	-066-				.004		.037
0.0082				-067-	-068-	-069-	-070-	-071-	-072-				.005		.045
0.01				-073-	-074-	-075-	-076-	-077-	-078-				.006		.055
0.012				-079-	-080-	-081-	-082-	-083-	-084-				.007		.066
0.015				-085-	-086-	-087-	-088-	-089-	-090-				.008		.082
0.018				-091-	-092-	-093-	-094-	-095-	-096-				.010		.100
0.02				-097-	-098-	-099-	-100-	-101-	-102-				.011		.110
0.022				-103-	-104-	-105-	-106-	-107-	-108-				.012		.121
0.027	0.562 (14.27)		0.762 (19.35)	-109-	-110-	-111-	-112-	-113-	-114-				.015		.149
0.033				-115-	-116-	-117-	-118-	-119-	-120-		.002		.018		.182
0.039				-301-	-302-	-303-	-304-	-305-	-306-		.002		.021		.215
0.047				-121-	-122-	-123-	-124-	-125-	-126-		.003		.026		.259
0.05				-127-	-128-	-129-	-130-	-131-	-132-		.003		.028		.275
0.056				-307-	-308-	-309-	-310-	-311-	-312-		.003		.031		.310
0.068	0.687 (17.45)		0.887 (22.53)	-133-	-134-	-135-	-136-	-137-	-138-		.004		.040		.374
0.082				-139-	-140-	-141-	-142-	-143-	-144-		.005		.050		.451
0.1		0.193 (4.90)		-145-	-146-	-147-	-148-	-149-	-150-		.006		.055		.550
0.12				-151-	-152-	-153-	-154-	-155-	-156-		.007		.066		.661
0.15	0.562 (14.27)	0.235 (5.97)	0.762 (19.35)	-157-	-158-	-159-	-160-	-161-	-162-		.008		.082		.824
0.20	0.687 (17.45)		0.887 (22.53)	-163-	-164-	-165-	-166-	-167-	-168-		.011		.110	20.0	1.000
0.22				-169-	-170-	-171-	-172-	-173-	-174-		.012		.121	18.5	1.017
0.27				-175-	-176-	-177-	-178-	-179-	-180-		.015		.149	17.0	1.148
0.33				-181-	-182-	-183-	-184-	-185-	-186-		.018		.182	16.0	1.322
0.39				-313-	-314-	-315-	-316-	-317-	-318-		.021		.215	15.0	1.463
0.47		0.312 (7.92)		-187-	-188-	-189-	-190-	-191-	-192-		.026		.259	14	1.645
0.50				-193-	-194-	-195-	-196-	-197-	-198-		.028		.275	13.5	1.688
0.68				-199-	-200-	-201-	-202-	-203-	-204-		.037		.374	12	2.041
0.82	0.813 (20.65)		1.013 (25.73)	-205-	-206-	-207-	-208-	-209-	-210-		.045		.451	10	2.049
1.0				-211-	-212-	-213-	-214-	-215-	-216-		.055		.550	8.4	2.100
1.5		0.400 (10.16)		-217-	-218-	-219-	-220-	-221-	-222-		.082		.824	5.8	2.172
2.0				-223-	-224-	-225-	-226-	-227-	-226-		.110		1.100	4.4	2.200
2.2				-229-	-230-	-231-	-232-	-233-	-234-		.121		1.210	4	
3.0	1.063 (27.00)		1.263 (32.08)	-235-	-236-	-237-	-238-	-239-	-240-		.165		1.654	2.9	
3.3			1.263 (32.08)	-241-	-242-	-243-	-244-	-245-	-246-		.182		1.818	2.7	
4.0	1.375 (34.93)		1.575 (40.01)	-247-	-248-	-249-	-250-	-251-	-252-		.220		2.200	2.2	
4.7		0.500 (12.70)		-253-	-254-	-255-	-256-	-257-	-258-		.259	18.7		1.87	
5.0				-259-	-260-	-261-	-262-	-263-	-264-		.275	17.6		1.76	
6.8		0.562 (14.27)		-319-	-320-	-321-	-322-	-323-	-324-		.374	13.0		1.30	
8.0				-325-	-326-	-327-	-328-	-329-	-330-		.440	11.0		1.10	
10.0				-331-	-332-	-333-	-334-	-335-	-336-		.550	8.8		.88	
15.0	1.875 (47.63)		2.075 (52.71)	-337-	-338-	-339-	-340-	-341-	-342-		.824	6.6		.66	
20.0		0.670 (17.02)		-343-	-344-	-345-	-346-	-347-	-348-		1.100	4.4		.44	
22.0				-349-	-350-	-351-	-352-	-353-	-354-		1.210	4.0		.40	

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 20261021T18:15Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical characteristics, dimensions, and dash numbers - Continued.

Capacitance value (nom) (in µF)	Dimensions 1/ (in inches with mm in parentheses) L ±.030 (.76)	D +.020 (.51) -.000	G, max	Dash number 2/ Capacitance tolerance value (in %) ±0.25	±0.5	±1.0	±2.0	±5.0	±10.0	AC ratings (for sinusoidal operation from -55° to +100°C) 3/ 0 to 4 KHz Volts	0 to 4 KHz Current (in A)	At 4 KHz Volts	At 4 KHz Current (in A)	At 40 KHz Volts	At 40 KHz Current (in A)
STYLES CHR01D, CHR01E, AND CHR01F - 50 VOLTS (DC RATING)															
0.001	0.500 (12.70)	0.170 (4.32)	0.700 (17.78)	-001-	-002-	-003-	-004-	-005-	-006-	36.0	.001	36.0	.001	36.0	.009
0.0012				-007-	-008-	-009-	-010-	-011-	-012-				.001		.012
0.0015				-013-	-014-	-015-	-016-	-017-	-018-				.001		.014
0.0018				-019-	-020-	-021-	-022-	-023-	-024-				.002		.016
0.002				-025-	-026-	-027-	-028-	-029-	-030-				.002		.018
0.0022				-031-	-032-	-033-	-034-	-035-	-036-				.002		.020
0.0027				-037-	-038-	-039-	-040-	-041-	-042-				.002		.020
0.0033				-043-	-044-	-045-	-046-	-047-	-048-				.003		.030
0.0047				-049-	-050-	-051-	-052-	-053-	-054-				.004		.042
0.005				-055-	-056-	-057-	-058-	-059-	-060-				.005		.045
0.0066				-061-	-062-	-063-	-064-	-065-	-066-				.006		.061
0.0082	0.562 (14.27)		0.762 (19.35)	-067-	-068-	-069-	-070-	-071-	-072-				.007		.074
0.01				-073-	-074-	-075-	-076-	-077-	-078-				.009		.090
0.012				-079-	-080-	-081-	-082-	-083-	-084-				.011		.108
0.015				-085-	-086-	-087-	-088-	-089-	-090-				.013		.135
0.018		0.193 (4.90)		-091-	-092-	-093-	-094-	-095-	-096-		.002		.016		.163
0.02				-097-	-098-	-099-	-100-	-101-	-102-		.002		.018		.180
0.022				-103-	-104-	-105-	-106-	-107-	-108-		.002		.020		.121
0.027				-109-	-110-	-111-	-112-	-113-	-114-		.002		.024		.243
0.033				-115-	-116-	-117-	-118-	-119-	-120-		.003		.030		.298
0.039	0.687 (17.45)	0.170 (4.32)	0.887 (22.53)	-283-	-284-	-285-	-286-	-287-	-288-		.004		.035		.351
0.047	0.687 (17.45)		0.887 (22.53)	-121-	-122-	-123-	-124-	-125-	-126-		.004		.040		.423
0.05				-127-	-128-	-129-	-130-	-131-	-132-		.005		.045		.450
0.056		0.193 (4.90)		-289-	-290-	-291-	-292-	-293-	-294-		.005		.050		.504
0.068				-133-	-134-	-135-	-136-	-137-	-138-		.006		.060	34.0	.578
0.082	0.813 (20.65)		1.013 (25.73)	-139-	-140-	-141-	-142-	-143-	-144-		.007		.070	32.0	.656
0.1				-145-	-146-	-147-	-148-	-149-	-150-		.009		.090	30.0	.750
0.12	0.687 (17.45)	0.235 (5.97)	0.887 (22.53)	-151-	-152-	-153-	-154-	-155-	-156-		.011		.108	30.0	.901
0.15				-157-	-158-	-159-	-160-	-161-	-162-		.013		.135	26.0	.974
0.20	0.813 (20.65)		1.013 (25.73)	-163-	-164-	-165-	-166-	-167-	-168-		.018		.180	24.0	1.200
0.22				-169-	-170-	-171-	-172-	-173-	-174-		.020		.198	20.0	1.100
0.27	0.687 (17.45)	0.312 (7.92)	0.887 (22.53)	-175-	-176-	-177-	-178-	-179-	-180-		.024		.243	19.0	1.282
0.33				-181-	-182-	-183-	-184-	-185-	-186-		.030		.298	18.0	1.488
0.39	0.813 (20.65)		1.013 (25.73)	-295-	-296-	-297-	-298-	-299-	-300-		.035		.351	17.0	1.658
0.47	0.813 (20.65)		1.013 (25.73)	-187-	-188-	-189-	-190-	-191-	-192-		.042		.423	16.0	1.880
0.50		0.400 (10.16)		-193-	-194-	-195-	-196-	-197-	-198-		.045		.450	15.0	1.875
0.68				-199-	-200-	-201-	-202-	-203-	-204-		.062		.612	14.0	2.381
0.82	1.063 (27.00)		1.263 (32.08)	-205-	-206-	-207-	208-	-209-	-210-		.074		.738	12.0	2.459
1.0				-211-	-212-	-213-	-214-	-215-	-216-		.090		.900	10.0	2.500
1.5				-217-	-218-	-219-	-220-	-221-	-222-		.135		1.348	8.0	2.996
2.0	1.125 (28.58)	0.500 (12.70)	1.325 (33.66)	-223-	-224-	-225-	-226-	-227-	-228-		.180		1.800	6.0	3.000
2.2				-229-	-230-	-231-	-232-	-233-	-234-		.198		1.980	5.8	3.190
3.0	1.375 (34.93)		1.575 (40.01)	-235-	-236-	-237-	-238-	-239-	-240-		.271		2.707	4.5	3.384
3.3				-241-	-242-	-243-	-244-	-245-	-246-		.298		2.975	4.1	3.389
4.0		0.562 (14.27)		-301-	-302-	-303-	-304-	-305-	-306-		.360		3.600	3.6	3.600
4.7		0.670 (17.02)		-307-	-308-	-309-	-310-	-311-	-312-		.423	30.0		3.2	
5.0				-313-	-314-	-315-	-316-	-317-	-318-		.450	28.8		3.0	
6.8	1.875 (47.63)		2.075 (52.71)	-319-	-320-	-321-	-322-	-323-	-324-		.612	21.2		2.2	
8.0				-325-	-326-	-327-	-328-	-329-	-330-		.720	18.0		1.8	
10.0				-331-	-332-	-333-	-334-	-335-	-336-		.900	14.4		1.5	

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 202610-21T18:15Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical characteristics, dimensions, and dash numbers - Continued.

Capacitance value (nom) (in µF)	Dimensions 1/ (in inches with mm in parentheses) L ±.030 (.76)	D +.020 (.51) -.000	G, max	Dash number 2/ Capacitance tolerance value (in %) ±0.25	±0.5	±1.0	±2.0	±5.0	±10.0	AC ratings (for sinusoidal operation from -55° to +100°C) 3/ 0 to 4 KHz Volts	Current (in A)	At 4 KHz Volts	Current (in A)	At 40 KHz Volts	Current (in A)
STYLES CHR01G, CHR01H, AND CHR01J - 100 VOLTS (DC RATING)															
0.001	0.500 (12.70)	0.170 (4.32)	0.700 (17.78)	-001-	-002-	-003-	-004-	-005-	-006-	60.0	.001	60.0	.002	60.0	.015
0.0012				-007-	-008-	-009-	-010-	-011-	-012-				.002		.018
0.0015				-013-	-014-	-015-	-016-	-017-	-018-				.002		.022
0.0018				-019-	-020-	-021-	-022-	-023-	-024-				.003		.027
0.002				-025-	-026-	-027-	-028-	-029-	-030-				.003		.030
0.0022				-031-	-032-	-033-	-034-	-035-	-036-				.003		.033
0.0027				-037-	-038-	-039-	-040-	-041-	-042-				.004		.041
0.0033				-043-	-044-	-045-	-046-	-047-	-048-				.005		.050
0.0047				-049-	-050-	-051-	-052-	-053-	-054-				.007		.071
0.005				-055-	-056-	-057-	-058-	-059-	-060-				.008		.075
0.0068	0.687 (17.45)		0.887 (22.53)	-061-	-062-	-063-	-064-	-065-	-066-				.010		.102
0.0082				-067-	-068-	-069-	-070-	-071-	-072-				.012		.123
0.01				-073-	-074-	-075-	-076-	-077-	-078-		.002		.015		.150
0.012				-079-	-080-	-081-	-082-	-083-	-084-		.002		.018	59.0	.177
0.015				-085-	-086-	-087-	-088-	-089-	-090-		.002		.022	58.0	.217
0.018		0.193 (4.90)		-091-	-092-	-093-	-094-	-095-	-096-		.003		.027	57.0	.257
0.02				-097-	-098-	-099-	-100-	-101-	-102-		.003		.030	55.0	.275
0.022				-103-	-104-	-105-	-106-	-107-	-108-		.003		.033	53.0	.292
0.027				-109-	-110-	-111-	-112-	-113-	-114-		.004		.041	51.0	.344
0.033				-115-	-116-	-117-	-118-	-119-	-120-		.005		.049	50.0	.413
0.039		0.235 (5.97)		-121-	-122-	-123-	-124-	-125-	-126-		.006		.059	48.0	.468
0.047				-127-	-128-	-129-	-130-	-131-	-132-		.007		.071	47.0	.552
0.050				-133-	-134-	-135-	-136-	-137-	-138-		.008		.075	46.0	.575
0.056				-139-	-140-	-141-	-142-	-143-	-144-		.008		.084	46.0	.644
0.068	0.813 (20.65)		1.013 (25.73)	-145-	-146-	-147-	-148-	-149-	-150-		.010		.102	42.0	.714
0.082	0.687 (17.45)	0.312 (7.92)	0.887 (22.53)	-151-	-152-	-153-	-154-	-155-	-156-		.012		.123	38.0	.779
0.10				-157-	-158-	-159-	-160-	-161-	-162-		.015		.150	36.0	.900
0.12				-163-	-164-	-165-	-166-	-167-	-168-		.018		.180	35.0	1.051
0.15	0.813 (20.65)		1.013 (25.73)	-169-	-170-	-171-	-172-	-173-	-174-		.022		.225	31.0	1.160
0.20				-175-	-176-	-177-	-178-	-179-	-180-		.030		.300	29.0	1.450
0.22				-181-	-182-	-183-	-184-	-185-	-186-		.033		.330	25.0	1.375
0.27	1.063 (27.00)		1.263 (32.08)	-187-	-188-	-189-	-190-	-191-	-192-		.041		.405	24.0	1.621
0.33				-193-	-194-	-195-	-196-	-197-	-198-		.050		.496	23.0	1.901
0.39		0.400 (10.16)		-199-	-200-	-201-	-202-	-203-	-204-		.059		.585	22.0	2.145
0.47				-205-	-206-	-207-	-208-	-209-	-210-		.071		.705	21.0	2.468
0.5				-211-	-212-	-213-	-214-	-215-	-216-		.075		.750	20.0	2.500
0.68	1.125 (28.58)	0.500 (12.70)	1.325 (33.66)	-217-	-218-	-219-	-220-	-221-	-222-		.102		1.020	16.0	2.721
0.82				-223-	-224-	-225-	-226-	-227-	-228-		.123		1.223	14.0	2.869
1.00		0.562 (14.72)		-229-	-230-	-231-	-232-	-233-	-234-		.150		1.500	12.0	3.000
1.50	1.375 (34.93)		1.575 (40.01)	-235-	-236-	-237-	-238-	-239-	-240-		.225	57.0	2.135	10.0	3.745
2.0		0.670 (17.02)		-241-	-242-	-243-	-244-	-245-	-246-		.300	55.0	2.750	8.1	4.050
2.2				-247-	-248-	-249-	-250-	-251-	-252-		.330	54.0	2.970	7.5	4.121
3.0	1.875 (47.63)		2.075 (52.71)	-253-	-254-	-255-	-256-	-257-	-258-		.451	52.0	3.910	6.0	4.511
3.3		0.670 (17.02)		-259-	-260-	-261-	-262-	-263-	-264-		.496	50.0	4.132	5.5	4.545
4.0		0.750 (19.05)		-265-	-266-	-267-	-268-	-269-	-270-		.600	48.0	4.800	4.8	4.800
4.7				-271-	-272-	-273-	-274-	-275-	-276-		.705	43.0	5.000	4.3	5.000
5.0				-277-	-278-	-279-	-280-	-281-	-282-		.750	40.0		4.0	
6.8	2.375 (60.33)	1.000 (25.40)	2.575 (65.41)	-283-	-284-	-285-	-286-	-287-	-288-		1.020	30.0		3.0	
8.0				-289-	-290-	-291-	-292-	-293-	-294-		1.200	24.0		2.4	
10.0				-295-	-296-	-297-	-298-	-299-	-300-		1.500	20.0		2.0	

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 20260-21T18:15Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical characteristics, dimensions, and dash numbers - Continued.

Capacitance value (mfd) (in µF)	Dimensions 1/ (In inches with mm in parentheses) L ±.030 (.76)	D ±.020 (.51) -.000	G, max	Dash number 2/ Capacitance tolerance value (in %) ±0.25	±0.5	±1.0	±2.0	±5.0	±10.0	AC ratings (for sinusoidal operation from -55° to +100°C) 3/ 0 to .4 KHz Volts	0 to .4 KHz Current (In A)	At 4 KHz Volts	At 4 KHz Current (In A)	At 40 KHz Volts	At 40 KHz Current (In A)
STYLES CHR01K, CHR01L, AND CHR01M - 200 VOLTS (DC RATING)															
0.001	0.562 (14.27)	0.170 (4.32)	0.762 (19.35)	-001-	-002-	-003-	-004-	-005-	-006-	120.0	.001	120.0	.003	80.0	.020
0.0012				-007-	-008-	-009-	-010-	-011-	-012-				.004		.024
0.0015				-013-	-014-	-015-	-016-	-017-	-018-				.005		.029
0.0018				-019-	-020-	-021-	-022-	-023-	-024-				.005		.036
0.002				-025-	-026-	-027-	-028-	-029-	-030-				.006		.040
0.0022				-031-	-032-	-033-	-034-	-035-	-036-				.007		.044
0.0027				-037-	-038-	-039-	-040-	-041-	-042-				.008		.054
0.0033				-043-	-044-	-045-	-046-	-047-	-048-				.010		.066
0.0039				-049-	-050-	-051-	-052-	-053-	-054-				.012		.070
0.0047				-055-	-056-	-057-	-058-	-059-	-060-				.014		.094
0.005				-061-	-062-	-063-	-064-	-065-	-066-		.002		.015		.100
0.0056				-067-	-068-	-069-	-070-	-071-	-072-		.002		.017		.112
0.0068				-073-	-074-	-075-	-076-	-077-	-078-		.002		.020		.136
0.0082				-079-	-080-	-081-	-082-	-083-	-084-		.002		.025		.164
0.01		0.193 (4.90)		-085-	-086-	-087-	-088-	-089-	-090-		.003		.030		.200
0.012				-091-	-092-	-093-	-094-	-095-	-096-		.004		.036	78.0	.234
0.015		0.235 (5.97)		-097-	-098-	-099-	-100-	-101-	-102-		.004		.045	76.0	.285
0.018	0.687 (17.45)		0.887 (22.53)	-103-	-104-	-105-	-106-	-107-	-108-		.005		.054	74.0	.333
0.02				-109-	-110-	-111-	-112-	-113-	-114-		.006		.060	71.0	.355
0.022				-115-	-116-	-117-	-118-	-119-	-120-		.007		.066	68.0	.374
0.027		0.312 (7.92)		-121-	-122-	-123-	-124-	-125-	-126-		.008		.081	65.0	.439
0.033		0.312 (7.92)		-127-	-128-	-129-	-130-	-131-	-132-		.010		.099	62.0	.512
0.039				-133-	-134-	-135-	-136-	-137-	-138-		.012		.118	60.0	.585
0.047				-139-	-140-	-141-	-142-	-143-	-144-		.014		.141	57.0	.670
0.050	0.813 (20.65)		1.013 (25.73)	-145-	-146-	-147-	-148-	-149-	-150-		.015		.150	56.0	.700
0.056				-151-	-152-	-153-	-154-	-155-	-156-		.017		.168	56.0	.784
0.068				-157-	-158-	-159-	-160-	-161-	-162-		.020		.204	50.0	.850
0.082				-163-	-164-	-165-	-166-	-167-	-168-		.025		.246	44.0	.902
0.1				-169-	-170-	-171-	-172-	-173-	-174-		.030		.300	42.0	1.050
0.12	0.813 (20.65)		1.013 (25.73)	-175-	-176-	-177-	-178-	-179-	-180-		.036		.360	40.0	1.201
0.15	1.063 (27.00)	0.400 (10.16)	1.263 (32.08)	-181-	-182-	-183-	-184-	-185-	-186-		.045		.449	36.0	1.348
0.18	1.375 (34.93)		1.575 (40.01)	-187-	-188-	-189-	-190-	-191-	-192-		.060		.600	34.0	1.700
0.22				-193-	-194-	-195-	-196-	-197-	-198-		.066		.660	32.0	1.760
0.27		0.500 (12.70)		-199-	-200-	-201-	-202-	-203-	-204-		.081		.810	29.0	1.958
0.33				-205-	-206-	-207-	-208-	-209-	-210-		.099		.992	28.0	2.314
0.39				-211-	-212-	-213-	-214-	-215-	-216-		.117		1.170	27.0	2.634
0.47				-217-	-218-	-219-	-220-	-221-	-222-		.141		1.410	26.0	3.055
0.5				-223-	-224-	-225-	-226-	-227-	-228-		.150		1.500	25.0	3.125
0.56		0.500 (12.70)		-229-	-230-	-231-	-232-	-233-	-234-		.170	115.0	1.580	23.0	3.165
0.68		0.562 (14.27)		-235-	-236-	-237-	-238-	-239-	-240-		.204	110.0	1.871	20.0	3.401
0.82	1.875 (47.63)		2.075 (52.71)	-241-	-242-	-243-	-244-	-245-	-246-		.246	100.0	2.049	19.0	3.893
1.0				-247-	-248-	-249-	-250-	-251-	-252-		.300	90.0	2.250	15.0	3.750
1.25				-253-	-254-	-255-	-256-	-257-	-258-		.370	87.0	2.701	13.0	4.050
1.5		0.670 (17.02)		-259-	-260-	-261-	-262-	-263-	-264-		.449	85.0	3.184	12.0	4.494
2.0		0.750 (19.05)		-265-	-266-	-267-	-268-	-269-	-270-		.600	81.0	4.050	10.0	5.000
2.5				-271-	-272-	-273-	-274-	-275-	-276-		.750	09.0	4.320	8.2	5.125
3.0		1.000 (25.40)		-277-	-278-	-279-	-280-	-281-	-282-		.902	67.0	5.038	7.0	5.283
4.0	2.375 (60.33)		2.575 (65.41)	-283-	-284-	-285-	-286-	-287-	-288-		1.200	54.0	5.400	5.4	5.400

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:15Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical characteristics, dimensions, and dash numbers - Continued.

Capacitance value (nom) (in µF)	Dimensions 1/ (in inches with mm in parentheses) L ±.030 (.76)	D +.020 (.51) -.000	G, max	Dash number 2/ Capacitance tolerance value (in %) ±0.25	±0.5	±1.0	±2.0	±5.0	±10.0	AC ratings (for sinusoidal operation from -55° to +100°C) 3/ 0 to .4 KHz Volts	0 to .4 KHz Current (in A)	At 4 KHz Volts	At 4 KHz Current (in A)	At 40 KHz Volts	At 40 KHz Current (in A)
STYLES CHR01N, CHR01P, AND CHR01R - 400 VOLTS (DC RATING)															
0.001	0.563 (14.27)	0.193 (4.90)	0.762 (19.35)	-001	-002	-003	-004	-005	-006	240.0	.001	240.0	.006	100.0	.025
0.002				-007	-008	-009	-010	-011	-012		.001		.012		.050
0.005	0.687 (17.45)	0.235 (5.97)	0.887 (22.53)	-013	-014	-015	-016	-017	-018		.003		.030		.125
0.01				-019	-020	-021	-022	-023	-024		.006		.060		.250
0.015	0.813 (20.65)		1.013 (25.73)	-025	-026	-027	-028	-029	-030		.009		.090	94.0	.352
0.022		0.312 (7.92)		-031	-032	-033	-034	-035	-036		.013		.132	85.0	.468
0.027				-037	-038	-039	-040	-041	-042		.016		.163	81.0	.547
0.033				-043	-044	-045	-046	-047	-048		.020		.198	78.0	.645
0.039				-049	-050	-051	-052	-053	-054		.023		.234	75.0	.731
0.047		0.400 (10.16)		-055	-056	-057	-058	-059	-060		.028		.282	71.0	.834
0.05				-061	-062	-063	-064	-065	-066		.030		.300	70.0	.875
0.068				-067	-068	-069	-070	-071	-072		.041		.408	60.0	1.020
0.082	1.063 (27.00)		1.263 (32.08)	-073	-074	-075	-076	-077	-078		.049		.492	50.0	1.025
0.10				-079	-080	-081	-082	-083	-084		.060		.600	46.0	1.150
0.15	1.375 (34.93)		1.575 (40.01)	-085	-086	-087	-088	-089	-090		.082	220.0	.824	40.0	1.496
0.22		0.500 (12.70)		-091	-092	-093	-094	-095	-096		.132	200.0	1.100	35.0	1.925
0.33		0.562 (14.27)		-097	-098	-099	-100	-101	-102		.198	180.0	1.322	32.0	2.843
0.47	1.875 (47.63)		2.075 (52.71)	-103	-104	-105	-106	-107	-108		.282	163.0	1.680	31.0	3.843
0.50				-109	-110	-111	-112	-113	-114		.300	160.0	1.750	30.0	3.750
1.0		0.750 (19.05)		-115	-116	-117	-118	-119	-120		.600	120.0	3.000	18.0	4.750
1.5	2.375 (60.33)	1.000 (25.40)	2.575 (65.41)	-121	-122	-123	-124	-125	-126		.899	110.0	4.120	16.0	5.893
2.0				-127	-128	-129	-130	-131	-132		1.200	100.0	5.000	14.0	7.000

1/ Dimensions are bare case sizes (see figure)
2/ The complete dash number will include the applicable letter completing the style designator (A, B, C, D, E, F, G, H, J, K, L, M, N, P, or R) as a prefix and the applicable FR level symbol as a suffix.
3/ For +125°C operation, linearly derate +100°C rating by 50 percent.

TABLE II. Measurement values.

Measurement	CHR01A, CHR01D, CHR01G, CHR01K, and CHR01N	CHR01B, CHR01E, CHR01H, CHR01L, and CHR01P	CHR01C, CHR01F, CHR01J, CHR01M, and CHR01R
Insulation resistance (terminal to terminal)			
Megohms x microfarads (min):			
At +25° C	300,000	200,000	100,000
At +100° C	30,000	20,000	5,000
At +125° C	2,000	1,000	1,000
Megohms (need not exceed):			
At +25° C	3,000,000	2,000,000	1,000,000
At +100° C	300,000	200,000	50,000
At +125° C	20,000	10,000	10,000
Cap. change with temp (max):			
At -55° C	-1.5% to -0.5%	-1.7% to -0.5%	-2.0% to -0.5%
At +25° C (retrace)	0.03%	0.05%	0.25%
At +125° C	+0.8 to -0.5%	+1.2% to -0.7%	±2.0%
Dissipation factor (max)	0.15%	0.2%	0.25%

Source: https://assist.dla.mil -- Downloaded: 20261O-21T18:15Z
Check the source to verify that this is the currentversion before use.

REQUIREMENTS - Continued.

Flash point of impregnant - Not applicable.

Radiographic inspection (X-ray) - Not applicable.

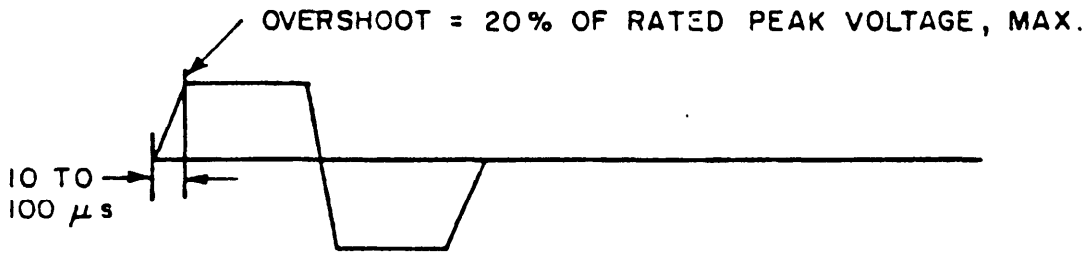
Ⓒ Thermal shock - Method 107 of MIL-STD-202, condition B.

Seal - Method 112 of MIL-STD-202, condition C, procedure IIIA; no evidence of leakage.
Leakage-rate sensitivity:
CHR01A, CHR01B, CHR01D, CHR01E, CHR01G, CHR01H, CHR01K, CHR01L, CHR01N, and CHR01P: 10^{-6} atm cc/sec.
CHR01C, CHR01F, CHR01J, CHR01M, and CHR01R: 10^{-5} atm cc/sec.

Ⓒ Failure rate level "L": At 25° C, immerse in isopropyl alcohol; check for continous bubble stream.

Dielectric withstanding voltage (DWV) - Method 301 of MIL-STD-202; no momentary breakdown.
Terminal to terminal:

Ⓒ AC: 100 ±10 Hz square wave, peak-to-peak voltage, three times dc rated voltage for 60 seconds, min. Need not exceed 800 V peak-to-peak.



DC: 200 percent of dc rated voltage for 60 seconds, min.

Insulation resistance (IR) - Method 302 of MIL-STD-202. Charge time 5 minutes, max; however, for cap. values greater than 1.0 μF, an additional 1 minute per μF is permitted.
Terminal to terminal (dc rated voltage): See table II.
Terminals to case: 10,000 megohms, min.

Cap. - Method 305 of MIL-STD-202.
Test frequency: 1 KHz ±100 Hz.
Limit of accuracy: Within ±0.05 percent.

Dissipation factor (DF) - See table II.
Test frequency: 1 KHz ±100 Hz (cap. values ≤ 1.0 μF) or 100 ±10 Hz (cap. values > 1.0 μF).
Limit of accuracy: Within ±0.05 percent or within 10 percent of table II value.

Barometric pressure (reduced) - Method 105 of MIL-STD-202, condition D (100,000 feet).
125 percent of dc rated voltage applied. See MIL-C-39022 for voltage limitations.

Vibration, high frequency - Method 204 of MIL-STD-202, condition D (20G).
50 percent of dc rated voltage applied.

Salt spray (corrosion) - Method 101 of MIL-STD-202, condition B (48 hours).
Salt solution: 5 percent.

Source: https://assist.dla.mil -- Downloaded: 20261 0-21T18:15Z
Check the source to verify that this is the current version before use.

REQUIREMENTS - Continued.

Immersion - Method 104 of MIL-STD-202, condition C.
DWV:
Insulating sleeves: Greater than 4,000 volts, dc.
Terminal to terminal: 200 percent of dc rated voltage.
Terminals to case: 200 percent of dc rated voltage.
IR:
Insulating sleeves: 1,000 megohms, min.
Terminal to terminal: Not less than 50 percent of value specified in table II.
Terminals to case: 5,000 megohms, min.
ΔCap.:
CHR01A, CHR01D, CHR01G, CHR01K, and CHR01N: ±0.25 percent, max.
CHR01B, CHR01E, CHR01H, CHR01L, and CHR01P: ±0.5 percent, max.
CHR01C, CHR01F, CHR01J, CHR01M, and CHR01R: ±1.0 percent, max.
DF: See table II.

Solderability - Method 208 of MIL-STD-202.

Shock (specified pulse) - Method 213 of MIL-STD-202, condition I.
50 percent of dc rated voltage applied. No arcing or other evidence of electrical breakdown.

Moisture resistance - Method 106 of MIL-STD-202.
DWV, IR, ΔCap., and DF: Same as for immersion.

Terminal strength - Method 211 of MIL-STD-202, conditions A and D.
Condition A: Applied force 5 pounds.
Condition D: 3 rotations of 360 degrees.

Low temperature and cap. change with temperature -
Low temperature: In accordance with MIL-C-39022.
Cap. change with temperature: In accordance with MIL-C-39022, except measurements may be made at -50 +0, -3°C, -25 ±3°C, 0 ±3°C, +25 ±3°C, +50 ±3°C, +75 ±3°C, +100 ±3°C, and +125 ±3°C.
ΔCap. (at -55°, +25° (retrace), and +125°C): See table II.

Fault count - Not applicable.

Fungus - Method 508 of MIL-STD-810.

Resistance to soldering heat - Method 210 of MIL-STD-202, condition B (260 ±5°C for 10 ±1 seconds).
IR: See table II.
ΔCap.: Within ±0.25 percent of initial value.
DF: See table II.

Resistance to solvents - Method 215 of MIL-STD-202.

Life (at +100°C) - Method 108 of MIL-STD-202.
(C) Accelerated conditions: 140 percent of dc rated voltage for 2,000 +72, -0 hours.
Rated conditions: 100 percent of dc rated voltage for 10,000 +96, -0 hours.
DF (at +100°C) between 24 and 48 hours of test - See table II.
IR -
Insulating sleeves: 1,000 megohms, min.
Terminal to terminal: Not less than 33.3 percent of initial requirement.
Terminals to case: 5,000 megohms, min.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:15Z
Check the source to verify that this is the current version before use.

REQUIREMENTS - Continued.

Life - Continued.
ΔCap. -
CHR01A, CHR01D, CHR01G, CHR01K, and CHR01N: ±2.0 percent of initial value.
CHR01B, CHR01E, CHR01H, CHR01L, and CHR01P: ±4.0 percent of initial value.
CHR01C, CHR01F, CHR01J, CHR01M, and CHR01R: ±5.0 percent of initial value.
DF (at +25° C) after life -
CHR01A, CHR01B, CHR01D, CHR01E, CHR01G, CHR01H, CHR01K, CHR01L, CHR01N, and CHR01P: 0.25 percent, max.
CHR01C, CHR01F, CHR01J, CHR01M, and CHR01R: 0.35 percent, max.
(C) AC life - 140 percent of ac rated voltage for 240 +48, -0 hours. Test samples shall be in accordance with table III.

Marking - In accordance with MIL-C-39022, and the following:
Part number: Consists of the basic number of this specification sheet with a dash number coded as shown in table I.

(C) EXAMPLE OF MARKING:

M39022/09-A001LJAN - Part Number and "JAN" Marking.
7304AB - Date Code and Lot Symbol.
TM XXXXX - Trademark and Source Code.

TABLE III. AC life test samples.

DC rated voltage (volts)	AC rating at 400 Hz (volts)	Life test voltage RMS	Life test voltage frequency	Total cap. value of sample (μF)	Cap. range for selection of sample
30	22	30	4 KHz	40	.001 to 4 μF
50	36	50	4 KHz	40	.001 to 4 μF
100	60	84	4 KHz	20	.001 to 2.2 μF
200	120	168	400 Hz	40	.001 to 10 μF
400	240	336	400 Hz	20	.001 to 2 μF

NOTES:

Cross-reference of substitute items - In accordance with table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:15Z
Check the source to verify that this is the current version before use.

TABLE IV. Cross reference of substitute items.

Superseded P/N	Replacement P/N	Superseded P/N	Replacement P/N	Superseded P/N	Replacement P/N
STYLES CHR01A, CHR01B, AND CHR01C - 30 VOLTS					
-265-	-319-	-277-	-331-	-289-	-343-
-266-	-320-	-278-	-332-	-290-	-344-
-267-	-321-	-279-	-333-	-291-	-345-
-268-	-322-	-280-	-334-	-292-	-346-
-269-	-323-	-281-	-335-	-293-	-347-
-270-	-324-	-282-	-336-	-294-	-348-
-271-	-325-	-283-	-337-	-295-	-349-
-272-	-326-	-284-	-338-	-296-	-350-
-273-	-327-	-285-	-339-	-297-	-351-
-274-	-328-	-286-	-340-	-298-	-352-
-275-	-329-	-287-	-341-	-299-	-353-
-276-	-330-	-288-	-342-	-300-	-354-
STYLES CHR01D, CHR01E, AND CHR01F - 50 VOLTS					
-247-	-301-	-259-	-313-	-271-	-325-
-248-	-302-	-260-	-314-	-272-	-326-
-249-	-303-	-261-	-315-	-273-	-327-
-250-	-304-	-262-	-316-	-274-	-328-
-251-	-305-	-263-	-317-	-275-	-329-
-252-	-306-	-264-	-318-	-276-	-330-
-253-	-307-	-265-	-319-	-277-	-331-
-254-	-308-	-266-	-320-	-278-	-332-
-255-	-309-	-267-	-321-	-279-	-333-
-256-	-310-	-268-	-322-	-280-	-334-
-257-	-311-	-269-	-323-	-281-	-335-
-258-	-312-	-270-	-324-	-282-	-336-

Preparing activity:
Air Force - 11

Agent:
DSA - ES

(Project 5910-F229-1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:15Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1 ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1426 1 OCT 76

Replaces edition of 1 Jan 72 which may be used.

S/N 0102-LF-001-4260

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:15Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE AIR FORCE

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 73236 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE AIR FORCE

4950/TZS
Wright-Patterson AFB, 45433

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-21T18:15Z
Check the source to verify that this is the currentversion before use.